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Offline and Preference-Based Reinforcement Learning for Robotics: A Survey

Abstract

Reinforcement learning (RL) has shown remarkable success in simulation and games, but real-world robotics applications pose significant challenges due to safety concerns, sample inefficiency, and difficulty in specifying reward functions. Two promising directions address these issues: **offline reinforcement learning**, which learns policies from pre-collected data without risky exploration, and **preference-based (human-in-the-loop) reinforcement learning**, which infers objectives from human feedback instead of hand-designed rewards. This survey provides a comprehensive overview of these approaches as applied to robotics. We first review offline RL methods — including policy-constraint, value-regularization, and model-based techniques — and discuss how they exploit logged robot datasets and visual pre-training to learn control policies safely and efficiently【20†L288-L293】【23†L19-L23】. Next, we survey preference-based RL, covering preference elicitation, reward learning from comparisons, and interactive algorithms that train robots with human guidance【13†L142-L149】【62†L72-L78】. In each area, we highlight key algorithms and their adaptation to robotic tasks (e.g. goal-conditioned offline RL【10†L9-L15】, video pre-training【30†L37-L45】, and sample-efficient preference learning【40†L30-L38】). We also identify current challenges — such as distribution shift, representation learning from high-dimensional sensor data, human feedback noise, and evaluation protocols — and prospects for future research. Our goal is to synthesize over decades of work and point out open problems in deploying offline and preference-based RL on real robots.

Introduction

Reinforcement learning (RL) enables agents to learn complex behaviors via trial and error, but its direct application to real-world robots is often impractical. Traditional **online RL** requires millions of environment interactions to train a policy, which in robotics means extensive real-world trials that are expensive, slow, and unsafe. Moreover, manually designing a dense reward function that captures desired behaviors is difficult and error-prone【62†L72-L78】【34†L153-L161】. To address these issues, two research directions have gained prominence:

- **Offline Reinforcement Learning (Offline RL)**: Offline RL (also called batch RL) learns entirely from previously collected datasets, without any further interaction with the environment【20†L288-L293】【23†L19-L23】. This allows leveraging large-scale logs of robot demonstrations or operations, and avoids costly or dangerous exploration. For example, robot control systems can be trained from datasets of logged trajectories (e.g. manipulation trials or legged locomotion)【20†L288-L293】【23†L19-L23】. Effective offline RL algorithms enable policy learning under distribution shift, making RL feasible in domains like robotics where online data collection is limited【20†L288-L293】【23†L19-L23】.

- **Preference-Based Reinforcement Learning (PbRL)**: PbRL replaces numeric reward engineering with human-provided preference feedback, allowing non-experts to guide learning via comparisons【13†L142-L149】【62†L72-L78】. Instead of crafting a reward for a task (e.g. how a door-opening skill should be scored), a human can simply compare two robot behaviors and indicate which is preferable【62 † L72-L78】【13 † L142-L149】. This paradigm has the advantages of producing dense, interpretable reward models aligned with human intent【62 † L72-L78】【13 † L142-L149】, and mitigating "reward hacking" that arises from misspecified objectives. Preference learning has been successfully applied in robotics to adapt policies to user-provided criteria (e.g. style or safety preferences) with far fewer trials than manual reward tuning【62 † L72-L78】【40 † L30-L38】.

In summary, offline RL aims to use data efficiently by avoiding online exploration, while preference-based RL aims to specify objectives flexibly through human feedback. Both approaches are particularly appealing for robotics: robots often have abundant logged data but limited ability to explore safely【20 † L288-L293】【23 † L19-L23】, and their tasks are often best defined by human preferences rather than ad-hoc rewards【62 † L72-L78】【13 † L142-L149】. This survey provides a unified treatment of these two topics in the context of robotic learning. We review fundamental methods, organize them into coherent families, and discuss how they have been or can be applied to robotics problems. We also highlight common challenges (such as distribution shift, representation learning, and human feedback efficiency) and promising future directions.

The remainder of this paper is organized as follows. Section 2 surveys offline RL methods, categorizing techniques like policy constraints, value-function regularization, model-based learning, and representation learning. Section 3 covers preference-based RL methods, including reward modeling from comparisons, interactive preference elicitation, and integration with policy optimization. Section 4 discusses challenges and prospects, focusing on issues in robotics applications and outlining open problems. Finally, Section 5 concludes. Throughout, we emphasize examples and evaluations relevant to robotic control and manipulation tasks, citing key works and known benchmarks.

Offline Reinforcement Learning Methods

Offline RL studies how to learn a policy from a fixed dataset $D = \{(s, a, r, s')\}$ collected by one or more behavior policies. In the robotics context, D might come from human demonstrations, teleoperated control logs, or self-generated experience. The key technical challenge is **distributional shift**: the learned policy may query actions or states not well-covered by D, causing overestimation of value and poor performance【36 † L69-L78】【47 † L27-L34】. Modern offline RL methods mitigate this via policy constraints or value pessimism, which we now review in turn.

Policy-Constraint and Behavior-Regularized Methods

A common strategy is to constrain the learned policy to stay close to the data distribution. For example, **Batch-Constrained Q-learning (BCQ)** uses a generative model of the behavior policy to restrict actions to those seen in the dataset【36 † L69-L78】. Fujimoto et al. (2019) introduced BCQ, which combines a learned action decoder with Q-learning: it generates candidate actions similar to the data and selects the highest-valued among them【36 † L69-L78】. By construction, BCQ never evaluates actions outside the support of D, thus avoiding extrapolation error and enabling learning from arbitrary offline data【36 † L69-L78】.

Related methods include **Bootstrapping Error Accumulation Reduction (BEAR)** (Kumar et al., 2019) and **Advantage-Weighted Actor-Critic (AWAC)** (Nair et al., 2020). BEAR penalizes the learned policy's divergence from the behavior data distribution, while AWAC performs policy updates weighted by advantages under the behavior policy. These approaches encourage the new policy to mimic the dataset so that training remains on known state-action pairs. In practice, simple variants often work well: for instance, **TD3+BC** (Fujimoto & Nachum, 2021) adds a behavior-cloning term to the TD3 algorithm, yielding a strong baseline for continuous control. More generally, the **Behavior Regularized Actor-Critic (BRAC)** framework (Wu et al., 2019) unifies several such methods via a regularization term that penalizes divergence from the dataset policy【49†L33-L39】. Wu et al. find that many recent offline RL methods can be seen as special cases of BRAC, and that even relatively simple constraints often suffice for good performance【49†L33-L39】.

A variant of policy constraint approaches uses trust-region ideas or explicit policy interpolation. For example, some methods learn a mixture of the behavior and learned policy, or impose KL-divergence penalties to keep them close. These are in spirit similar to behavior regularization. In robotics applications, policy-constraint methods have enabled learning control from suboptimal data: Chebotar et al. (2021) leverage a goal-conditioned version of BCQ to perform unsupervised skill discovery from offline datasets of robotic trajectories【10†L9-L15】. Their **Actionable Models** algorithm learns to reach any past goal state by relabeling goals and training a BCQ-style agent, demonstrating that robots can acquire diverse manipulation skills from offline logs without explicit rewards【10†L9-L15】.

Value-Pessimism and Regularization Methods

Another approach is to penalize value estimates on unseen actions. **Conservative Q-Learning (CQL)** (Kumar et al., 2020) explicitly adds a regularizer to the Q-function update that lowers the values of out-of-distribution actions【47†L27-L34】. CQL aims to learn a "conservative" Q-function whose expected value lower-bounds the true value under the dataset distribution【47†L27-L34】. The result is a lower-bounded estimate of policy performance, which improves robustness to distribution shift. Kumar et al. show that CQL substantially outperforms prior offline RL methods on various benchmarks【47†L27-L34】. In practice, CQL augments the Bellman error loss with a term that reduces the Q-values of unvisited actions, thereby avoiding optimistic overestimation.

Similarly, **Critic Regularized Regression (CRR)** and **Conservative Expectile Regression (IQL)** take pessimistic views on value learning. The recent **Implicit Q-Learning (IQL)** (Kostrikov et al., 2022) avoids querying unseen actions by fitting the state-value via an upper expectile of the return distribution【59†L45-L52】. IQL alternates between estimating the optimal state value (via expectile regression) and performing policy extraction by advantage-weighted imitation. This method never directly evaluates unknown actions, yet it can still learn policies better than the behavioral baseline. IQL achieves state-of-the-art results on the D4RL benchmark suite【59†L45-L52】, illustrating the power of implicit value pessimism【59†L45-L52】.

Theoretical work has also emphasized pessimism in offline RL: frameworks like **penalized MDPs** formalize adding penalties proportional to model uncertainty or distributional shift【53†L35-L43】. For instance, MOPO (Yu et al., 2020) and MOReL (Kidambi et al., 2020) learn dynamics models and construct a "pessimistic MDP" that lower-bounds the true returns【53†L35-L43】【55†L24-L33】. MOPO penalizes model rollout rewards by uncertainty, and MOReL builds an MDP whose transition model abstains in uncertain regions. These methods theoretically guarantee that the learned policy's true return is close to its estimated return in the penalized model. Empirically, MOPO and MOReL outperform standard offline baselines on continuous control tasks, especially when generalization to new states is needed【53†L35-L43】【55†L24-L33】.

More recently, Yu et al. (2021) introduced **COMBO**, which sidesteps explicit uncertainty estimation by directly regularizing value on out-of-support transitions【57†L28-L36】. COMBO trains a value function using both the offline data and model-generated rollouts, then enforces a conservative value estimate for any state-action pair that is not in the dataset【57†L28-L36】. This yields a value function that is provably pessimistic outside the data manifold, yielding safe policies without requiring explicit uncertainty quantification【57†L28-L36】. Experiments show that COMBO matches or exceeds prior offline RL methods on benchmarks and especially excels at tasks demanding generalization beyond the training data【57 † L33-L39】.

Model-Based Offline RL Methods

Model-based approaches attempt to explicitly model the robot's dynamics and use planning under uncertainty. Building on MOReL and MOPO, model-based offline RL algorithms typically proceed in two steps: (a) learn a dynamics model from D, and (b) plan or optimize a policy under a conservative version of the model. For example, **MOReL** first fits a model and constructs a pessimistic MDP that penalizes uncertain transitions【55 † L24-L33】, then solves it as a surrogate for the true environment. MOReL is shown to be minimax-optimal (up to log factors) and matches state-of-the-art on offline benchmarks【55 † L24-L33】. MOPO similarly uses model uncertainty to penalize rewards【53 † L35-L43】.

These model-based methods have been extended to handle high-dimensional observations common in robotics. For instance, Dreamer and other latent-space model-based algorithms have been adapted for offline visual control. Lu et al. (2023) explore offline RL from pixel observations, emphasizing robustness to visual distractions. They find that by combining simple modifications of existing model-based (Dreamer) and model-free (DrQ-v2) algorithms, one can outperform more complex baselines on continuous control with images【7 † L19-L27】. Such works illustrate that offline model-based RL can leverage large video and image datasets to train perceptual models for robotic tasks.

Pre-training from large-scale video datasets is another exciting avenue. Bhateja et al. (2023) propose **V-PTR**, which pre-trains a value function on internet-scale egocentric video (e.g. Ego4D) via temporal-difference learning【30 † L37-L45】. By learning rich representations from diverse human videos, V-PTR "primes" the robot's value estimator. When fine-tuned on diverse robot data, the resulting policies "perform better, act robustly, and generalize broadly" in manipulation tasks【30 † L37-L45】. In experiments on a WidowX robot, V-PTR outperforms prior offline RL methods on real manipulation tasks【30 † L37-L45】. This line of work demonstrates the synergy of offline RL with large vision-language datasets, enabling robots to benefit from "internet-scale" visual pre-training.

Representation and Goal-Conditioned Learning

A related thread is learning goal-conditioned or unsupervised skills from offline data. If the dataset contains diverse outcomes, one can treat each state or observation as a potential goal and learn a goal-reaching policy. Chebotar et al. (2021) do this by treating every state as a goal and applying Q-learning with hindsight relabeling【10 † L9-L15】. The agent thus learns how to reach any state seen in the data using offline updates. This approach, called **Actionable Models**, yields a wide repertoire of behaviors without manual reward labels. More generally, recent work on unsupervised RL and self-supervised learning (e.g. contrastive or latent dynamics models) can be incorporated into offline training to extract useful representations for robotics.

Offline RL in Robotics Benchmarks

To facilitate research, several benchmarks and datasets have been released. **D4RL** (Fu et al., 2020) provides offline datasets for continuous control (e.g. MuJoCo locomotion and manipulation) collected by controllers, human demonstrations, or replay buffers【61 † L29-L38】. These datasets reveal the deficiencies of simple offline RL baselines and have become a standard evaluation suite【61 † L29-L38】. In robotics, benchmarks like D4RL (with simulated tasks) and real-world datasets (e.g. surgical robot logs) test the viability of offline methods. Recent work has also introduced vision-based offline benchmarks and multi-task datasets to better reflect real robotic challenges【61 † L29-L38】. Overall, offline RL methods continue to improve on these benchmarks, but deploying them on physical robots still requires addressing challenges such as generalization and robustness【20†L288-L293】【23†L19-L23】.

Preference-Based Reinforcement Learning Methods

Preference-based RL (PbRL) encompasses a variety of methods where the learning signal is provided by comparisons or qualitative feedback rather than explicit numeric rewards. Formally, one assumes access to preferences over trajectories or behaviors (e.g. trajectory τ_1 is preferred to τ_2), and the goal is to learn a policy that maximizes the (unknown) underlying reward consistent with these preferences【13†L142-L149】【34†L153-L161】. PbRL can relieve the burden of reward design and can be applied even by non-experts who can judge which behavior they like better.

Learning from Pairwise Comparisons

A common PbRL framework learns a reward model that predicts human preferences, then optimizes a policy using that learned reward. Given a dataset of pairwise comparisons $\{(\tau_i^1, \tau_i^2)\}$ labeled by a human, one fits a function $R(s, a)$ such that the cumulative rewards of the preferred trajectories exceed the others. A popular choice is to model the probability that τ_1 is preferred to τ_2 using a logistic function of the reward sums, akin to the Bradley–Terry model. This was the approach of Christiano et al. (2017), who applied deep RL with human preferences to train agents on complex tasks【15†L29-L38】. Their work showed that robots and simulated agents can learn behaviors (like locomotion or manipulation) from human comparisons alone, without explicit reward functions【15†L29-L38】. Recent IRL approaches also interpret preference labels as weak supervision and infer a hidden reward that respects the orderings.

Learning to rank preferences effectively requires careful modeling. Some methods integrate uncertainty or Bayesian treatment of the reward model to handle noisy or inconsistent labels. Others use iterative or active querying to select trajectory pairs that are most informative to label (so-called active preference learning). For example, algorithms can generate candidate trajectories and ask the human to compare them, thus focusing feedback on uncertain parts of the policy space. Active methods draw from principles of experimental design to minimize the number of queries needed. In robotics, active preference learning is attractive because real users may only tolerate a few queries: e.g., a system might propose a few perturbed trajectories of a robot and ask which one the human likes better, thereby quickly tuning the behavior.

Once a reward model R is learned, one can treat it as a standard (though learned) reward function and apply RL. For instance, after training a neural network to predict preferences, one can plug it into an off-the-shelf RL algorithm (e.g. PPO or SAC) to optimize the robot policy. This two-stage approach has been used in many domains, including robotics, where humans provide preference labels on pairs of robot trajectories or videos. An example is training a mobile robot: the human watches two behavior videos (e.g. one where the robot follows behind a user at a certain distance vs. another distance) and picks which is preferred; the system updates its reward model and replans.

Direct Preference-Based Policy Search

Some PbRL algorithms skip explicit reward modeling and learn a policy directly from preferences. One classical example is **Policy Gradient from Human Preferences**, which treats the human comparisons as providing a policy-gradient-like feedback. Akrour et al. (2011) and Cheng et al. (2011) showed that preference data can be integrated into policy gradient or tree search methods【18†L63-L71】. More recently, end-to-end preference-based policy optimization methods have been proposed. These methods typically define a loss function comparing the probability of generating preferred vs. non-preferred trajectories. By maximizing this preference likelihood, the agent learns to generate trajectories that humans will rank higher. For instance, Tambe et al. (2023) propose Behavior Preference Regression (BPR) for offline RL, which directly regresses a policy to maximize the likelihood of observed trajectory rankings, and they demonstrate it can improve over offline baselines【18†L63-L71】. Such direct approaches avoid training an intermediate reward and can incorporate constraints that ensure new policies align with human data.

Human-in-the-Loop RL and Real Robots

In practice, PbRL often involves an interactive loop with a human trainer. The agent alternately generates behaviors (trajectories, episodes) and queries the human for feedback. This feedback can be pairwise preferences, but can also include demonstrations, critiques, or evaluations (numeric or ordinal). Well-known frameworks include TAMER【64†L1-L4】 and COACH, where humans give real-time signals about the robot's actions. These can be viewed as special cases of PbRL with rich feedback signals. For example, Knox & Stone (2009) showed that scalar approval/disapproval signals can effectively shape a robot's behavior.

In contemporary robotics, PbRL has been used for style adaptation: adjusting a pre-trained robot policy to human preferences. For instance, Marta et al. (2025) address style adaptation by learning a low-rank adaptation to a reward model so that a mobile robot follows a user on the right side rather than the left. They show that naive fine-tuning leads to catastrophic forgetting of the original task, while their **FLoRA** method efficiently incorporates new preference data without losing prior skills【42†L39-L48】. Similarly, Liu et al. (2023) demonstrate that using a learned dynamics model in PbRL can dramatically reduce the number of required queries on real robots【40†L30-L38】. In their experiments, a quadruped or manipulator learns customized policies with human preferences by synthesizing safe queries in imagination, rather than risking the real robot.

Preference-Based Reward Learning and Hybrid Approaches

Preference data often has to be integrated with other forms of data. In robotics, one might combine human rankings with offline logs or sparse reward signals. For example, a robot might first train on an offline dataset with behavior cloning, then refine its policy from a few human comparisons. Some approaches also use semi-supervised learning: treating preference labels as partial supervision and propagating them via consistency or ranking losses. Others explore preference transfer, where preferences learned on one task guide another (e.g. learning invariant objectives across users).

A related thread is **Reward Learning from Critique or Advice**: humans might provide high-level advice or constraints instead of (or in addition to) pairwise labels. For example, one can encode preferences as logical constraints on the task (such as "keep the object level during pouring"), or as guidance from natural language. These richer forms of feedback lie on the spectrum between numeric rewards and pairwise comparisons. They can be integrated into the PbRL framework by converting them into soft constraints on the learned reward or policy. Research in human-robot interaction studies how different feedback modalities (preferences, demonstrations, corrections, language) can complement each other for teaching robots.

Evaluation Protocols

Evaluating PbRL methods requires considering both the policy performance and the human feedback efficiency. Common metrics include the number of human queries needed to reach a desired performance, and the final task performance under the learned policy. In practice, many works simulate preferences using "oracle" reward functions on benchmarks such as MuJoCo locomotion or manipulation tasks, asking a virtual agent to provide comparisons. While simulation studies provide controlled comparisons (e.g. [41] shows that SEER drastically reduces feedback needs on standard tasks), real robot experiments are relatively rare. Notable examples include learning from actual user comparisons on real robot arms or mobile robots (e.g. Marta et al., [42]). There is a growing need for standardized benchmarks for PbRL in robotics, possibly involving human evaluation of learned behaviors or realistic human feedback simulators.

Challenges and Prospects

Despite rapid progress, significant challenges remain in applying offline and preference-based RL to robotics:

- **Distributional Shift and Safety**: Offline RL must contend with the fact that the learned policy may select out-of-distribution actions. In robotics, this can have safety implications: a policy that mistakes an unobserved action for high reward might execute dangerous motions. Techniques like conservative value estimation and policy constraints help mitigate this【47 † L27-L34】【57 † L28-L36】, but principled guarantees are limited. Evaluating robustness under disturbances or adversarial perturbations is also critical. Ayabe et al. (2025) examine robustness of offline RL policies on simulated robots and find that offline policies often lack the ability to recover from unexpected state perturbations【23 † L37-L41】. Real robots require even more stringent safety measures. Future work should combine offline RL with formal safety constraints or risk-aware optimization to ensure learned policies do not take hazardous actions.

- **Representation Learning and High-Dimensional Sensing**: Robot tasks frequently involve high-dimensional inputs (vision, depth, tactile), but offline RL datasets may not cover all visual variations. Generalizing from pixel observations remains difficult. Works like Bhateja et al.[30] and Lu et al.[7] begin to address this by pre-training on large vision datasets, but more research is needed on representation learning that is robust to domain shifts. Self-supervised pre-training (contrastive learning, predictive modeling) on offline robot data could help, as could leveraging language or semantic information to structure the representation space.

- **Limited and Noisy Feedback**: In PbRL, humans provide relatively sparse feedback. Ensuring that few preference queries suffice is challenging, especially for complex tasks. Methods like SEER [41] and FLoRA [42] improve data-efficiency, but current systems still often require hundreds of comparisons for good results. Humans can be inconsistent or biased in their preferences, which can mislead learning. Handling noisy or partially contradictory feedback requires robust preference models and possibly interfacing with active learning to detect and correct errors. User interface design is also important: queries should be understandable and preferences intuitive. Semi-supervised approaches that combine a few human labels with much unlabeled or automated experience may alleviate this bottleneck.

- **Evaluation and Benchmarks**: Unlike offline RL (where D4RL provides a benchmark suite), there is no standard benchmark for preference-based RL in robotics. Existing studies often use toy tasks or simulation with synthetic preference oracles. Creating realistic benchmarks (even with simulated human experts or video data) would accelerate progress. Likewise, robotics tasks often require success metrics beyond scalar reward (e.g. task completion, safety constraints), and translating human preferences into these metrics is non-trivial.

- **Multi-Task and Transfer Learning**: Robots typically face many tasks or variations. Offline RL from multi-task datasets and PbRL that transfer preferences across tasks are promising directions. Meta-learning approaches could allow a robot to quickly adapt to a new user’ s preferences by leveraging previous users’ data. Few-shot PbRL (as in [62]) is one step in this direction, but more work is needed on meta-reward-learning and multi-user settings.

- **Integration with Other Learning Signals**: In practical robotics, diverse feedback may be available: some sparse rewards, some demonstrations, some preferences, and plenty of sensor data. Algorithms that can integrate offline data, demonstrations, and preferences coherently are needed. For instance, one could pre-train a reward model from offline logs using inverse RL or contrastive methods, then refine it with a few preference labels. Hybrid methods that combine imitation learning with preference-based fine-tuning may yield robust performance even from suboptimal datasets.

- **Algorithmic Advances**: On the algorithm side, improving uncertainty quantification, representation of preference models, and optimization stability remains open. Recent successes in language RL (RLHF) suggest that aligning policies with human values via large models can scale, but robotics has unique challenges (continuous control, real-time constraints). Techniques from robust RL, distributional RL, and Bayesian inference may help.

Looking forward, we expect offline and preference-based RL to become increasingly important for robotics. The availability of large robotic datasets (internet video of humans, multi-robot demonstrations, etc.) and advances in simulation mean that pretrained models will play a larger role. We foresee a synergy between offline learning from rich multimodal data and human preference fine-tuning: robots might first learn general capabilities (e.g. grasping, navigation) from logs and then adapt to specific user needs via a few preference queries. Safety will require combining these methods with formal verification or fail-safe control. Finally, interdisciplinary work involving HRI (human-robot interaction) will be crucial to design feedback protocols that are natural for humans and effective for learning.

Conclusion

Offline reinforcement learning and preference-based RL offer complementary solutions to key obstacles in robot learning. Offline RL enables leveraging historical data to train control policies without further risky exploration, while preference-based RL allows objectives to be specified interactively by humans. In this survey, we have reviewed the main algorithmic families for each approach and discussed their applicability to robotics tasks. We have highlighted how methods such as policy constraints (BCQ), value pessimism (CQL, IQL), model-based planning (MOReL, COMBO), and representation learning (video pre-training) can empower robots to learn from static datasets【36†L69-L78】【30†L37-L45】. We have also covered how human preferences can be used to shape robot behavior via reward modeling and direct policy learning【62†L72-L78】【40†L30-L38】. Despite promising results, many open challenges remain in scalability, human feedback efficiency, and safety. We hope this survey provides a solid foundation for researchers and practitioners to advance the state of the art in data-efficient, human-aligned robot learning.

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Note: References [23–50] above cover additional background and related methods (e.g. classical IRL, deep learning, meta-learning) to provide context. All references are to peer-reviewed publications or arXiv preprints in machine learning and robotics.
